6.5 Exponential Growth and Decay

Name _____ Period ____ Date _____

Algebra I 6.5 · Student sheet

I can…

- identify and write exponential growth and decay functions.
- find the percent rate of change of an exponential function.
- solve real-life problems, including compound interest.

Vocabulary · fill in the missing word

Exponential growth — $y = a(1 + r)^t$: a quantity _____ by the same percent each time period

Exponential decay — $y = a(1 - r)^t$: a quantity _____ by the same percent each time period

Growth / decay factor — _____

Compound interest — $y =$ _____

Part 1 · Viral post and phone battery

A post has 200 views, and its views grow by 50% each hour. A phone battery is at 100% and loses 10% of its charge each hour.

Hour	0	1	2	3	4	Multiply by	Equation
Views	200						
Battery (%)	100						

Why is the multiplier for a 50% increase 1.5 and not 0.5?

Part 2 · Notes and examples (fill in each note, then work the example)

Exponential growth: $y =$ _____ **Exponential decay:** $y =$ _____

Reading the model: a is the _____, and r is the _____ written as a decimal.

Example 1

For $y = 500(1.06)^t$, identify the initial amount and the rate of growth. Evaluate when $t = 3$.

Example 2

Write a function for each situation. Use t for the number of years.

a. A stock valued at $250 decreases in value by 5% each year.

b. A population of 1,200 grows by 3% each year.

Example 3

You buy a car in 2020 for $32,000. Its value decreases by 12% each year. Find its value in 2026, to the nearest dollar.

Example 4

Is the table exponential growth, exponential decay, or neither?

t: 0 1 2 3

y: 64 48 36 27

Growth and decay factors: The growth factor is _____; the decay factor is _____.

Example 5

Does each function show growth or decay? Find the percent rate of change.

a. $y = 50(0.8)^t$ b. $y = 10(1.035)^t$

Example 6

Rewrite $y = 0.5(1.2)^{2t}$ to find the percent rate of change per year.

Compound interest: y = _____

Example 7

A $9,000 deposit earns 6% annual interest compounded semiannually. Write a function for the balance and find it after 5 years.

Part 3 · Partner practice

Partner 1

A $2,000 laptop loses 20% of its value each year. Write a function and find its value after 3 years.

Partner 2

A $500 deposit earns 4% annual interest compounded quarterly. Write a function and find the balance after 2 years.